SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 15
Duration : 60 Minutes

Code of Conduct

I JASMITHA R(192524295) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: JASMITHA R

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Total Marks Awarded:

Assessment Tasks

MCQ Test

1. **A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?**

   a) 5 TB
   b) 10 TB
   c) 15 TB
   d) 20 TB

2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?

~4 minutes
b) ~40 minutes
c) ~4 hours
d) ~24 hours

3. A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?

a)5000
b) 6000
c) 7000
d) 10,000

4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?

1 hour
b) 10 hours
c) 1.1 hours
d) 5.5 hours

5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?

a) G~4 hours
b) ~8 hours
c) ~20 hours
d) ~40 hours

6. A cloud provider offers pay-as-you-go pricing at $0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?

a) $50
b) $500
c) $1000
d) $2000

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

a) 9.8 Gbps
b) 9.6 Gbps
c) 8.0 Gbps
d) 7.5 Gbps

8. A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

a) 91.25 GB
b) 85 GB
c)100 GB
d) 120 GB

9. A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

a) 500 Mbps
b) 250 Mbps
c) 200 Mbps
d) 1000 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

a) 262 minutes
b) 365 minutes
c) 525 minutes
d) 150 minutes

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center?

a) 4,000
b) 8,000
c) 6,000
d) 10,000

12. A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?
   a. 4 GB
   b) 6 GB
   c) -8 GB
   d) 10 GB

13. A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?
   a. 16
   b) 32
   c) 64
   d) 128

14. A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?
   a. 40 kW
   b) 50 kW
   c) 60 kW
   d) 70 kW

15. A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?
   a. 6,400
   b) 12,800
   c) 25,600
   d) 50,000

16. Which cloud deployment model is used for organizations that require both private and public cloud resources?
   A): Hybrid Cloud
   B): Public Cloud
   C): Private Cloud
   D): Community Cloud

17. Which cloud security strategy ensures that sensitive data remains private even if intercepted?

   A): Encryption
   B): Load balancing
   C): Virtualization
   D): Resource pooling

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

a: 5000 GB
b: 7000 GB
c: 6000 GB
d: 7500 GB

19. A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

a: 56 GHz
b: 58 GHz
c: 64 GHz
d: 70 GHz

20. If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

a: 1.5 sec
b: 2 sec
c: 1 sec
d: 3 sec

21. If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month.

a: 600 GB
b: 500 GB
c: 700 GB
d: 800 GB

22. A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)?

a: 21.6 min
b: 36 min
c: 43.2 min

d: 18 min

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

a: 120 MB
b: 100 MB
c: 150 MB
d: 200 MB

24. A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement.
a: 180
b: 150
c: 120
d: 160

25. A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total?

a: 1200
b: 1000
c: 800
d: 600

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?

a: 12 min
b: 10 min
c: 6 min
d: 8 min

27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.
Formula:
Energy = Units × Power × Time (hrs)
a: 420 kWh
b: 480 kWh

c: 400 kWh
d: 500 kWh

28. Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy?

a: 32 TB
b: 35 TB
c: 30 TB
d: 28 TB

29. Which of the following is NOT a benefit of virtualization?

i. Reduced hardware costs
b) Increased energy consumption
c) Improved resource management
d) Faster deployment of applications

30. Which feature in virtualization allows multiple OS environments to run simultaneously?

i. Hypervisor
b) Load balancer
c) Firewall
d) BIOS

I.

1. c) 15 TB $5 \times 3 = 15$

2. b) 99.99% Given,

$$= \frac{99.99}{\frac{\cancel{0.01}}{100}} \times 3\cancel{65} \times 24 \times 60$$

$$= \frac{0.01}{100} \times 8760$$

$= \sim 40$ minutes

3. b) $\frac{60000}{10} = 6000$

4. c) $500 \times 8 = 4000$

$$\frac{4000}{3600} = 1.1 \text{ hours}$$

5. a) Given,

99.95

$$= \frac{0.05}{100} \times 365 \times 24$$

$= 0.0005 \times 8760$

$= 4.38$

$= 4$ hrs

6. b) 10 TB = 10000

10000×0.05

$= 500$

7. $2\% = \frac{2}{100}$

10 Gbps

$10 - 2\% = 10 - 0.02$

a) = 9.8 Gbps

8. 250×365

a) 91.25 GB

9. 5×100

a) = 500 Mbps

10. 99.95

$= \frac{0.05}{100} \times 365 \times 24$

$= 0.0005 \times 8760$

$= 4.38 \times 60$

a) = 262.8

a) 262 minutes

11. 200×40

b) 8000

12. $\frac{128}{12} \times 10$

d) = 10 GB

13. 64×2

b) $= \frac{128}{4} = 32$

14. 60%
remaining 40%
$40\% \times 100 = 400$
a) 40 KW

15. $100 \times 64 \times 4$
a) 25600

16. a) Hybrid cloud

17. a) Encryption

18. $50 \times 20 = 1000$
per week = 7
1000×7
$= 7000$
b. 7000 GB

19. $32 \times 2.5 \times 70$
a) = 56 GHZ

20. a) 1.5 sec

21. a) 600GB

22. $= \frac{0.05}{100} \times 30 \times 24 \times 60$
a) = 21.6 min

23. a) $\frac{200}{10} \times 60 = 120$ M

a) 180

25. $20 \times 20 \times 3$

a) 1200

26. $2 \times 60 \times 10$

a) 12 min

27. $5 \times 3.5 \times 24$

a) 420 KWh

28. $20 \times \frac{20}{100} = 4 \Rightarrow 20 - 4 = 16 \times 2 = 32$

a) 32 TB

29. b) Increased energy consumption

30. a) Hypervisor.